HOMEWORK 2.7 · UNIT 2 · NC.8.EE.4 · NC.8.EE.3

Operations with Scientific Notation

Sep 16 / Sep 17 · Targets: EE.4-1, EE.4-2, EE.4-3, EE.3-1, EE.3-4, EE.3-7, EE.4-6, EE.4-7, EE.4-8

Name _____ Date _____ Block _____

Remember: Multiply: multiply the coefficients, **add** the exponents. **Divide:** divide the coefficients, **subtract** the exponents. Then fix the form: $1 \le$ coefficient < 10. Example: $(6 \times 10^4)(5 \times 10^2) = 30 \times 10^6 = 3 \times 10^7$.

SCORE 3.0 Approaching · the basics

Targets: EE.4-1, EE.4-2, EE.4-3, EE.3-1

1 Multiply. $(4 \times 10^6)(2 \times 10^3)$ EE.4-1

2 Divide. $(9 \times 10^8) \div (3 \times 10^2)$ EE.4-2

3 Multiply. Fix the form. EE.4-1,EE.4-3

$(5 \times 10^3)(4 \times 10^5)$

4 Divide. Which is the correct answer in scientific notation? $(1.2 \times 10^{-4}) \div (4 \times 10^{-9})$ EE.4-2,EE.4-3

Hint: Is the coefficient between 1 and 10?

(A) 3×10^4
(B) 0.3×10^5
(C) 3×10^{-13}
(D) 3×10^5

5 Write in scientific notation. $6{,}300{,}000$ REVIEW 2.6 EE.3-1

SCORE 4.0 Proficient · on target

Targets: EE.4-2, EE.3-4, EE.3-7, EE.4-6

6 A red blood cell is about 8×10^{-6} m wide. A grain of sand is about 6.4×10^{-4} m wide. How many times as wide is the sand? EE.4-2,EE.3-4

(A) 80 times
(B) 8 times
(C) 800 times
(D) 0.8 times

7 **Lena** made a mistake. Circle it. Explain it. Fix it. REVIEW 2.6 EE.3-7

LENA'S WORK
$0.0034 = 3.4 \times 10^{3}$

8 Simplify. $\dfrac{(7.5 \times 10^{-6})(4 \times 10^{3})}{2.5 \times 10^{-4}}$ EE.4-6

Hint: Do the top first.

SCORE 5.0 Mastery · go deeper

Targets: EE.4-7, EE.4-8

9 Each hair grows about 3.5×10^{-4} m a day. A person has about 1×10^{5} hairs. How many days until all the hairs together grow 7×10^{3} m? Justify each step. EE.4-7

In one day all the hairs grow _____ . Then I divide _____ . It takes _____ days.

10 A movie file is 4×10^{9} bytes. A phone holds 1.28×10^{11} bytes. Kai says the phone holds 3.2×10^{1} movies. Is that reasonable? Explain. EE.4-8

Kai is _____ because _____ .